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APPLIED RESEARCH

Hybrid MRMR-PCA BagDT–An Effective Feature Selection Based Ensemble Model for Real-Time Intrusion Detection in IoT Environment

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ABSTRACT The escalating number of Internet of Things devices driven by the surging popularity of smart devices has led to the accumulation of large quantities of personal data. While this data can be used to create more intelligent and efficient devices, it also raises concerns about data protection and privacy. Though there are a variety of security threats like man-in-the-middle, replay, remote hijacking, password guessing, Denial of Service, and malware attacks, Distributed DoS attacks play a vital role in the IoT environment. The reason is that DDoS attacks can be initialized by any intruder at a cheaper cost. In some parts of the world, some platforms support hackers unknowingly. DDoS attacks can affect the functionality of a smart solution and sometimes it can lead to life threats in the healthcare sector when an IoT wearable doesn't function at the expected time frame. The urge of the hour is to develop methodologies to adapt to a real-time environment and which react immediately when there is an intrusion. This paper focuses on developing a hybrid framework based on Ensemble Learning for detecting the DDoS attack types in a very short time frame and alerting the users. The model addresses key challenges identified in existing approaches, including model size, deployment issues, and detection speed. The proposed MRMR-PCA-based Bagged Decision Tree Ensemble Model was tested on the CICIoT2023 dataset, which contains twelve different types of DDoS attacks. The experimental results prove that this model has a boost in accuracy by 2.5%, a reduction in training time by 86%, and a spike in prediction speed by 48.28%. This enhances the security of sensitive data in cloud storage as well as during communication offering a powerful tool for rapid intrusion detection.

INDEX TERMS Attacks, ensemble learning, Internet of Things, intrusion detection system, principal component analysis, smart devices.

I. INTRODUCTION

A recently emerged technology called the Internet of Things (IoT) has taken a big leap in the real-time environment and also has become an inseparable part of our daily lives [11]. This technology has a great impact and has transformed cities across the globe into smart cities. A recent survey conducted reveals that the total number of smart devices which was around 8.6 billion during 2019 has risen to 9.76 billion during 2020 and has reached 11.28 billion in 2021. The survey also predicted that there will be around 29.42 billion

IoT-connected smart devices during the year 2030 [12]. The major goal of this smart environment is to connect as well as deploy billions of smart IoT devices for supporting various applications across different industrial sectors like agriculture [13], manufacturing factories [14], transportation [15], healthcare [16], Industrial Control Systems [17] etc. as illustrated in the Figure 1. The advancement in technology has paved the way for remote monitoring in a smarter way. To handle all the applications, huge data is being generated by these smart devices in real-time. The data gathered is too sensitive in information which motivates the intruders to attack these devices or the communication channel or the storage area. The recent world has witnessed numerous

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TABLE 1. State-of-art methods:contributions and research gaps.

Ref.	Dataset	Model Used	Attacks Identified	Accuracy	Research Gap
[1]	CICIoT2023	LSTM	Brute force attack Web based Spoofing Mirai Recon DoS	98.75%	Model size is high Deployment issues Heavy weight model Training time is higher
[2]	CICIDS2017 N-BaIoT CICIoT2023	DL-BiLSTM	Brute force attack Web based Spoofing Mirai Recon DoS	CICIoT2023: 93.13% CICIDS2017: 99.67% N-BaIoT : 99.9%	Incorporate model quantization Detection rate is higher
[3]	CICIoT2023 IoTID20	Blended counterfactual and Local Interpretable Model-Agnostic Explanations	DoS Scan Mirai	CICIoT2023: 99.51% IoTID20: 100%	Scalability issues Model selection is trivial Resource requirement is high
[4]	CICIoT2023 CICDDoS2019	modified Gated Recurrent Units	Multi-vector DDoS	CICIoT2023: 97.24% CICIoT2019: 97.19%	Assumption made on healthcare Authentication issues
[5]	CICIoT2023	Random Forest	Brute force attack Web based Spoofing Mirai Recon DoS	99.2%	Statistical technique for feature selection
[6]	CICIoT2023 TONIoT	Collaborative Intrusion Detection Framework	Brute force attack Web based Spoofing Mirai Recon DoS	CICIoT2023: 77.24% TONIoT: 99.99%	Lack of labelled heterogeneous data Black-box decision making
[7]	CICIoT2023	DNN CNN RNN	Brute force attack Web based Spoofing Mirai Recon DoS	RNN: 96.56%	Incremental training is required
[8]	CICIoT2023 NSL-KDD IDS2017 IDS2018 UNSW-NB15	Deep Reinforcement Learning Transferable Framework	Brute force attack Heartbleed Web Attack Botnet SSH DDoS	CICIoT2023:95.6% NSL-KDD:96% IDS2017:88% IDS2018:96% UNSW-NB15:94%	Speed of model training
[9]	CICIoT2023 Edge-IIoTset	Explainable Artificial Intelligence	Man in the middle injection attacks DDoS Ransomware	CICIoT2023:99% Edge-IIoTset: 92%	Time complexity of SHAP processes
[10]	CICIoT2023	ML Algorithms	Brute force attack Web based Spoofing Mirai Recon DoS	Improved by 3.72%	Feature selection required Class balancing required

cyberattacks [18] since all these devices are connected via the internet 24/7 and data is shared [19]. Hence, the need of the hour is to build efficient mechanisms [20] for securing the sensitive data generated and received by the smart IoT devices from the attackers [21], [22]. The mechanism that is currently suggested and is being used is to include a tier in the architecture of the IoT environment having Intrusion Detection Systems (IDS) [23]. The IDS is designed for monitoring the device as well as the network for any suspicious behaviour. When such behaviour is noticed, warnings are initiated to alert the users as well as the officials. This mechanism proactively tries to reduce

the impact of any intrusion which might be successful otherwise [24].

IDSs can be classified into five different types as discussed below. Network IDS are those security systems that are installed at a specific point of the network where it can monitor the traffic. Usually, these types of IDSs are installed on the subnets where firewalls are set up. The reason behind this is that these IDS will keep monitoring if anyone is trying to crack the firewall and alert the administrator on a viable threat. Host IDS are those which are installed in the individual IoT devices and they keep monitoring the packets moving in and out of the device. If any malicious

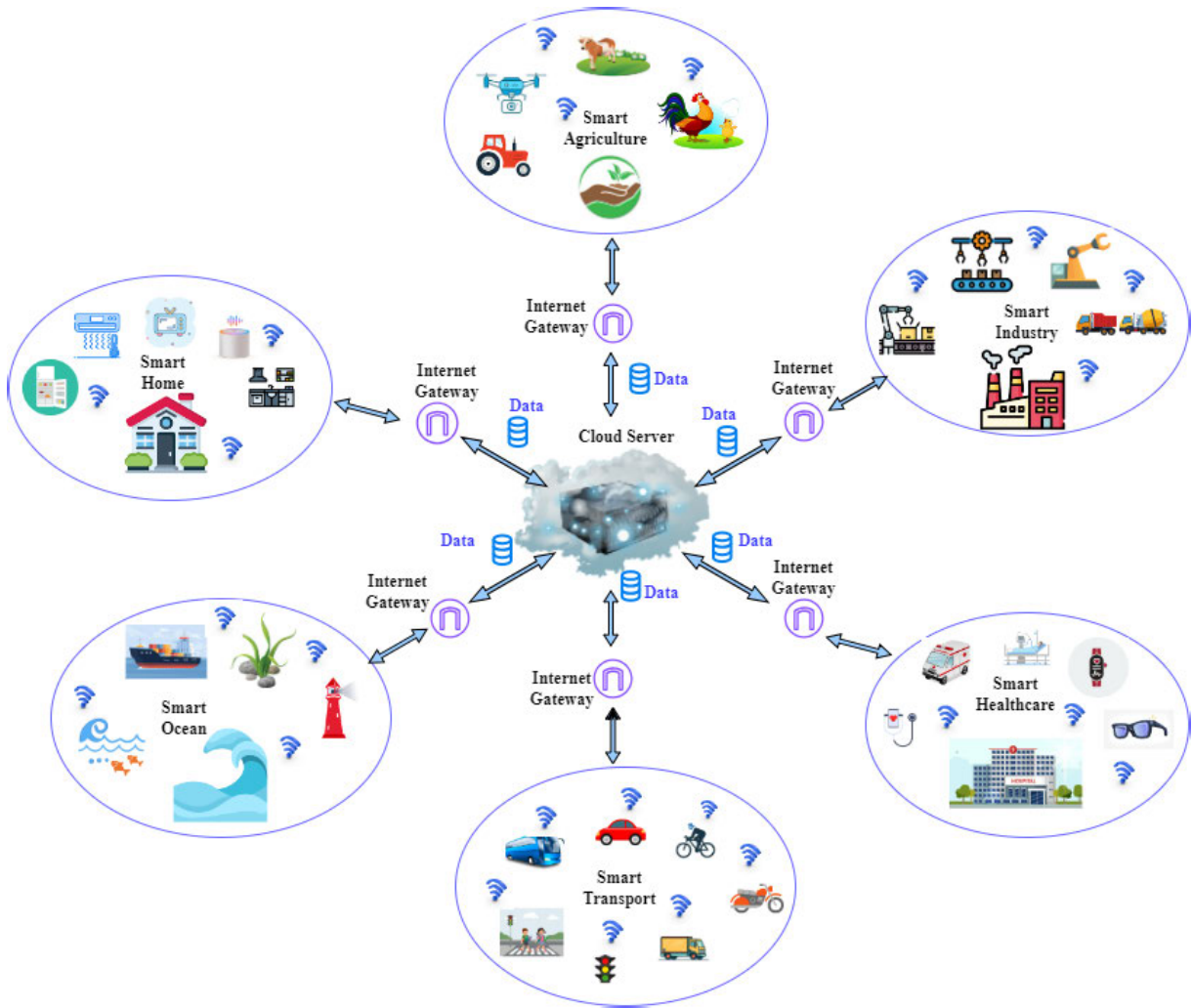


FIGURE 1. Illustration of smart environment in various applications.

packets are detected, the administrator is alerted. Protocol-based IDS are set up in the front end of the web server. The HTTPS protocol is regularly monitored by this type of IDS and tries to allow only authorized HTTP protocol requests. This helps in securing the web server from any attacks. Application Protocol-based IDS resides on a group of servers and continuously monitors the application-oriented protocols like SQL for securing the database. Hybrid IDS integrates any two or more methodologies discussed already for securing the complete network. Prelude is one such example of the hybrid IDS. This type of IDS is more efficient when compared to others. There are two methods by which IDS detects the attacks. They are Signature-based and Anomaly-based. The signature-based method tries to detect malicious activity by analyzing patterns like several 1s and 0s or bytes in the packets travelling via the network. Hence, these methods can be advantageous only when the patterns are well known, but when there is a new pattern of intrusion, they fail to detect. To overcome this drawback, the Anomaly-based methods were discovered. These methods use ML to build a model that detects suspicious packets based on its learning capability.

The recent decade has seen rigorous development in the field of IDS [24]. Though multiple methodologies are proposed by researchers, there are few limitations as illustrated in the table 1. In the case of IoT environment, since there are heterogeneous devices spread across the globe and the network via which data is shared is the Internet, it is necessary to develop a methodology for detecting the intrusion very fast, accurate and also compact so that IDS can be set up in multiple points to get a complete view of the smart environment. This paper proposes an IDS model for the IoT environment based on Ensemble Learning (EL). The major steps of this proposed model are pre-processing, feature selection and extraction, feature space dimension reduction and classification using the EL approach. The major contributions of this research and experimental validation are as follows.

- A hybrid methodology blending the Minimum Redundancy Maximum Relevance feature selection algorithm and Principal Component Analysis feature dimension reduction algorithm (MRMR-PCA) is proposed for improving the performance of the IDS by extracting only

high-impact features from the huge data generated by IoT devices.

- A Bagging-based Decision Tree Model is built using the EL approach which can handle class imbalance problems and also develops a very strong learner that can produce accurate results in real-time.
- The hybrid framework proposed can be instantly deployed in the real-time smart environment as the model size is very compact.
- Since the training time and model size are lesser and the prediction speed is higher, the proposed IDS model when adapted to a real-world IoT environment, the manufacturers will gain the trust of end users in using the smart IoT devices as their data will be highly secured.

The rest of the paper is organized as follows. Section II discusses about the related works, Section III deals with the background which includes Simple EL techniques and Advanced EL techniques. Section IV deals with materials and methods where the EL algorithms, Classifiers, Feature selection algorithm and dimension reduction algorithm for the features extracted are discussed. Section V discusses the proposed methodology for IDS in IoT environment. Section VI deals with the results and discussion on the results obtained. The paper is finally concluded with a performance analysis and future directions.

II. LITERATURE SURVEY

Many researchers have recently focused on developing appropriate methodologies and frameworks for building a secured IoT environment. This section discusses some recent methodologies developed by researchers across the world based on the CICIOT2023 real-time IoT data set [25]. When a survey is done on intrusion detection systems, the literature does not provide us with details regarding their research trend or the popularity of the dataset. Usually, most authors do not take research articles for review based on any qualitative measure. They take them arbitrarily. However, we focused more on the quality as well as the dataset being used. The evaluation criteria and performance were also taken into account. For qualitative measure, the citation was considered to be a measure. This metric can also be used as a quantitative measure of the popularity of the article and the author. Though there are various search engines, we used Google Scholar, Scopus, and Web of Science for gathering the articles. We considered articles published from 2017 to 2024. The search strings used are “Intrusion Detection Systems” and “Classification of Attacks in IoT Environment”. Then we excluded articles which were not related to the dataset CICIOT2023 [25]. Also, as the dataset we are focusing on was released in 2023, few authors had developed models for classifying these attacks. In [1], the authors have developed a methodology using long short-term memory for detecting various cyber-attacks currently available and those evolving. They claim an accuracy of 98.75% and F1 score of 98.59%. The LSTM model is utilized to adapt the dynamic changes in a real-time IoT environment. The major contribution of

the developed model is that it can respond very well to the dynamic nature of IoT data and accepts variable length sequences. The experiment was carried out using a batch of training data approximately 1000 in size. The testing was carried out for 50 epochs. Even though their model has high accuracy, they need to concentrate on model compression and quantization so that the model can be used in real time. Also, scalability issues need to be taken up while deploying in real-time. The authors in [3] tried to enhance the detection of attack using Local Interpretable Model-Agnostic Explanations. The proposed method in this research is a blending model which blends the three classifiers namely Gradient Boosting, Random Forest and Decision Tree. The final predicted result is based on the voting mechanism. Then the blended mode is evaluated using a learning curve and confusion matrix. The experimental results are validated using two data sets CICIOT2023 and IoTID20. They claim an accuracy of 99.51% for the CICIOT2023 data set and 100% for IoTID20 [26] data set. In [6], the authors have designed a collaborative intrusion detection framework namely CIDIoT for preserving privacy in IoT networks. The methodology uses an improved generative adversarial network model which do not exchange any network data. This model also efficiently handles the imbalance problems and communication overhead. This model can also handle the drawbacks of various traditional federated learning (FL) models when used for training. The author has also proposed a novel method for data aggregation using which the imbalance problems of the uploaded FL model is handled. The model was evaluated using two datasets namely CICIOT2023 and ToNIOT [27]. An accuracy of 98.9% was achieved when the CIDIoT model was experimented over the ToNIOT dataset whereas an accuracy of 77.20% was achieved when the CICIOT2023 dataset was used.

A novel approach for detecting multi-vector distributed DDos attacks is proposed which is an anomaly-based IDS [4]. Usually, anomaly-based IDS requires a very long processing time and are highly complex. Hence the authors tried to develop a lightweight anomaly-based IDS named modified Gated Recurrent Units (mGRU) for deploying in time-critical applications. The author claims that this model is well-suited for mobile healthcare systems. The experimental results are validated using the CICIOT2023 and CICDDoS2019 [28] data sets. A reduction of 2% in time consumption is achieved when mGRU is used. In [5] the authors have used PCA to eliminate the correlations among the features. Then for detecting the attacks, Random Forest is used giving an accuracy of 99.2%. The performance of the model is measured by having 5, 8, 10 and 47 features on the CICIOT2023 data set. Few researchers [7] applied Deep Neural Networks (DNN), Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) on CICIOT2023 data set for testing the efficacy. They inferred from experimental results that RNN performed best with an accuracy of 96.5%. An approach named transferable and adaptable Network IDS was proposed by [8] using Deep Reinforcement Learning.

TABLE 2. Comparison of proposed work with state-of-the-art techniques.

Ref.	Training Time	Prediction Speed	Model Size	Accuracy	Deployment	Light Weight	Feature Selection	Scalability
[1]	H	L	H	H	C	N	N	N
[2]	H	L	L	H	C	Y	Y	N
[3]	H	L	H	H	C	N	N	N
[4]	M	M	L	H	C	Y	Y	N
[5]	M	M	H	H	C	N	N	N
[6]	H	M	M	L	E	Y	N	N
[7]	H	M	M	H	E	N	Y	Y
[8]	H	M	M	H	E	N	N	Y
[9]	H	M	M	H	E	N	Y	N
[10]	M	M	M	H	E	N	N	N
Proposed	L	H	L	H	E	Y	Y	Y

H High M Medium L Low C Complex.
E Easy. N No. Y Yes.

The major focus was to prioritize the outliers and to transfer the features to the state of the environment. This paved the way for not needing any dimension for the feature space. Another set of researchers proposed a zero-trust Network IDS by integrating the Explainable AI technique [9] for detecting cyber-attacks in marine applications. This section discussed the various related works that were carried out recently using the data set CICIoT2023 for detecting intrusion in the IoT environment. Though many researchers have focused on developing efficient models for classifying the intrusion, every model has their own drawbacks. The major limitation in all these already proposed models is that they are not lightweight so they can be deployed in real-time. Also, the time required for training a heavy-weight classifier model is higher and the performance also depends on the performance of the processor on which these models run. Having this in focus, we have tried to develop a novel methodology for building a lightweight as well as a robust classifier that can detect the intrusion quicker with higher accuracy so that the post effects are lesser. A comparison of the advantages of the proposed model with all the state-of-the-art methodologies is illustrated in the table 2.

III. BACKGROUND

This section discusses in detail the limitations of traditional machine learning classifiers, the need for Ensemble Learning and the various state-of-the-art techniques currently available.

A. ENSEMBLE LEARNING

The recent decade has seen momentous success in the field of knowledge discovery due to tremendous development in Machine Learning (ML) and Artificial Intelligence

(AI). Though the traditional ML methods produce promising results, they sometimes fail to provide expected high-performance results when the data is complex. For example, when the data is completely imbalanced or noisy or multi-dimensional, the results obtained are not satisfactory. The major reason behind this unsatisfactory result is that these traditional ML models fail to capture numerous characteristics of the data and its structure. This imitation motivated the researchers to develop a methodology for effectively constructing a knowledge discovery model which resulted in Ensemble Learning (EL) [29].

EL is also a kind of ML technique that focuses on enhancing accuracy as well as flexibility by integrating various ML models' predictions. The major aim of the ensemble model is to lessen the total number of errors and biases that each ML model might have by integrating only their intelligence. The EL technique tries to combine the prediction outputs of diversified ML models and ends with a more precise and accurate output. Since varieties of ML models' strengths are focused majorly, EL improves the performance of the knowledge discovery machine.

Let us consider a real-time scenario to state how EL can be a better model when compared to traditional ML models. Assume that person X has developed a wearable device for diagnosing the physiological parameters of the body using various sensors. X wants to get feedback about his product before publishing it in the market. Consider that he asks his friend Y to provide feedback. Since Y is a close friend of X, he feels that he shouldn't disappoint or discourage X and gives positive feedback. Hence, now this feedback from Y is a biased one. To overcome this bias, let us consider that X approaches ten known neighbours but not close to him and asks for feedback. There is a chance that he will get unbiased feedback on his product. However, the problem behind this is that few neighbours may be aware of the need for the device or the working pattern of the same to provide an in-depth analysis. To get a more precise and accurate analysis, X can approach a few experts in the domain, some unknown personalities, some known personalities and land with precise feedback which might help him to improve his device so that it is competitive in the market. This is exactly what EL also does. Every ML model has its own advantages and disadvantages. EL tries to take diverse ML models that would perfectly suit the knowledge and combine decisions of these sets of ML models to land with a robust result. The working methodology of any EL model is that it uses various ML algorithms to produce a weak result. The predictive results are obtained based on the extracted features using projecting methods and finally, the results are combined using different voting mechanisms for achieving robust performance than when using an individual ML model.

B. SIMPLE ENSEMBLE TECHNIQUES

The simple techniques followed in EL are Max voting, Averaging and Weighted Average schemes [30].

TABLE 3. Ten point scale rating - on wearable device developed by X.

N1	N2	N3	N4	N5	N6	N7	N8	N9	N10
7	8	7	8	7	8	9	7	7	8

1) MAX VOTING SCHEME

The Max Voting scheme is a very simple technique that can be generally used for classification-based problems. As the name suggests, the one with maximum voting is considered to be the output. Assume that four classification ML models are used and each ML model produces a prediction result for the same set of data. Each ML model's prediction result is considered to be a vote. The prediction result which gets the maximum vote out of the ML models is considered to be the final result. Let us consider the scenario which was discussed in section III-A. If X gets the feedback from ten neighbours (N) on a 10-point scale rating and the individual rating is as shown in the table below, when a max voting scheme is followed, the final result on the scale will be 7. This scheme can be mathematically represented using the equation 1, where 'r' represents the predicted result by each ML model, 'X' represents the data for which prediction is required and 'R' represents the set of all predicted results.

$$pred(X) = \text{Max} - \text{Occurrence}(r \in R) \quad (1)$$

2) AVERAGING SCHEME

The averaging scheme is best suited for regression and for calculating the probabilities in the classification models. This is a simple averaging method similar to mean calculation in basic mathematics. Here, the predicted results of individual classifiers are noted down and the average of all the results is considered to be the final predicted result. In our scenario discussed already, the final feedback scale rating will be the average of all the ratings given by all ten neighbours. From the ratings provided in the table 3, if the averaging scheme is applied, the final result will be 7.6. The mathematical representation of this scheme can be as given in the equation 2, where 'X' represents the data for which prediction is to be performed, 'r' is the individual prediction results obtained, 'n' is the total number of ML models used for prediction.

$$pred(X) = \left(\sum_{i=1}^n r \right) \div n \quad (2)$$

3) WEIGHTED AVERAGING SCHEME

This scheme is an upgraded version of the averaging scheme. The ML models that are used for prediction are initially assigned a weight based on their importance level. The importance level can be based on the accuracy, rate of learning, expertise level, etc. For example, let us take the same scenario which was described in the previous section where X requests feedback from ten of his neighbours. Now, consider that out of the ten neighbours, four of them are experts in

developing wearable devices, four of them are well versed in the usage of wearable devices and the other two have no prior idea of this. In this case, the weight given for the first four will be higher than the next four. Whereas the last two will be assigned a very low weight as shown in the table 4 below. When the rating is calculated using this scheme, the final rating will be 7.59 which is a still precise prediction. This can be mathematically expressed as given in the equation 3, where 'w' is the weight corresponding to each prediction and 'r' is the individual prediction result obtained.

$$pred(X) = \left(\sum_{i=1}^n r * w \right) \div \left(\sum_{i=1}^n w \right) \quad (3)$$

C. ADVANCED ENSEMBLE TECHNIQUES

This section describes the various advanced techniques that are used for EL like Stacking, Blending, Bagging and Boosting [31].

1) STACKING TECHNIQUE

Stacking is a type of technique used in EL for improving the performance of the overall task of prediction by exploring the space of various basic models for the same problem. This method is also referred to as Stacked Generalization. In this technique, individual ML models' predictions are combined by using another model to improve the overall prediction performance. Here the individual model is termed as base learner and the model with which it is integrated is termed as the meta-learner or the blender. The steps involved in the stacking technique are as follows.

- The base learners can be multiple ML algorithms like Decision Trees, Random Forest, Support Vector Machines or Neural Networks or the same algorithm can be utilized with different hyper-parameters.
- The data is subdivided into various folds and a few folds are kept aside for validation. The base learners are used to train the remaining folds of data. Based on this learning performed, the prediction is tried out on the fold which was fixed for validation.
- The prediction results of the base learners are now used as features for training the meta-learner. The meta-learner will try to learn the knowledge by combining these predictions to make the final prediction.
- After the training is complete using the meta-learner, the test data is tested using the base-learner which produces a prediction. These predictions made by the base learner are then sent to the already trained meta-learner which finally predicts the class.

This EL technique can be used only when the dataset is huge enough to support the task of training with multiple ML models. The performance is better only when the diversity among the base learners is high. However, care should be taken to tune the models carefully so that over-fitting is avoided.

TABLE 4. Ten point scale rating - on wearable device developed by X along with weights.

Neighbour Count	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10
Weight Assigned	0.5	0.5	0.5	0.5	0.3	0.3	0.3	0.3	0.1	0.1
Scale	7	8	7	8	7	8	9	7	7	8

2) BLENDING TECHNIQUE

The Blending technique is similar to stacking. The major difference is that in stacking the entire data is subdivided into training and testing sets, and the training set is further subdivided into numerous folds whereas in blending only training and testing sets are taken for further processing. Hence, the blending technique uses a holdout concept and applies the base learner on the training set to perform prediction. Using the predictions achieved and the validation data, the new model is created considering them as features. Then using the new model created, the predictions are performed.

3) BOOT STRAP AGGREGATION TECHNIQUE

Bootstrap Aggregation is a method that is designed to improve the stability and accuracy of any ML algorithm that could be used in classification and regression problems. This methodology is sometimes referred to as Bagging. This technique reduces the variance and the over-fitting issue is handled effectively. Bagging is generally applied to Decision tree-based classification algorithms. This is an upgraded version of the model averaging scheme. The steps involved in the Bagging methodology are as follows.

- The original Data set “DS” is subdivided into several subsets “ss” each holding an equal number of elements “i” or tuples in it. The subsets can have unique tuples or can have a few duplicates as well.
- Each subset is learnt using a weak base model.
- The model is made to learn the knowledge of each subset paralleling but independently.
- The prediction result is equivalent to either the aggregated result based on a scoring or max-voting scheme.

4) BOOSTING TECHNIQUE

The Boosting technique can be used if the motive is to build a very strong classifier model using multiple weak classifier models. This method also has a high level of stability and reduces variance. The basic notion is to identify the misclassified data by one weak classifier and try to correct the mis-classification using a different classifier. The step-by-step procedure of the Boosting technique is given below.

- The complete data set “DS” is subdivided into training set “T” and testing set “t”.
- The data points “d” in the set “T” are assigned with weights “w”. Every “d” is assigned with the same “w” initially.
- Identify a weak classifier model and feed the data points “d” to it.

- Identify the data points “e” which are classified wrongly by the first weak model.
- Change the weights “w” of these data points to “w2” by increasing their magnitude. Also, reduce the magnitude of weights “w” of data points “d” to “w1” which were correctly classified by the first model.
- Another weak model is chosen and the set of data points “e” is predicted. The prediction error is corrected to some extent using this model.
- These steps are repeated until all the data points are classified correctly or till a maximum number of weak models are utilized.

IV. MATERIAL AND METHODS

This section discusses in detail the various algorithms and methods used for detecting the occurrence of attacks in the IoT environment. A detailed description of the various Ensemble learners and Classification learners as well as the feature selection algorithm used in this proposed methodology is discussed.

A. ENSEMBLE LEARNING METHODS

This subsection discusses in detail the mathematical intuition of the various ensemble methods used namely Adaptive Boosting Algorithm, RUSBoost Algorithm and Random Subspace Method.

1) ADAPTIVE BOOSTING ALGORITHM

Adaptive Boosting Algorithm [32] is popularly known as AdaBoost. The reason behind this name is that the weights assigned to the data points initially are reassigned to a higher value during every instance if they are misclassified. During the next iteration, these data points with higher weights are given more importance. The iterations are continued till the error is reduced. The AdaBoost algorithm follows the following steps.

- Consider that there is a data set ‘M’ which has totally ‘n’ data points. The initial weights for these data points are assigned using the equation 4 where ‘ w ’ is the weight assigned for each data point and ‘i’ represents the specific instance.

$$w(i) = 1/N \quad (4)$$

- A classifier C_j is chosen for performing the training and ‘j’ specifies the current iteration. The classifier which is chosen here has to at least have an accuracy greater than 0.5. This classifier C is considered to be the first stump.

- Once the classification is complete by using the classifier C_j on the data samples, the performance α of this stump is calculated using the equation 5. In this equation, the total error $ER(C_j)$ is calculated by summing up all the weights of the data points which are misclassified when using the current stump and are represented as $mW(i)$. This is illustrated in the equation 6.

$$\alpha = \frac{1}{2} \log_e ((1 - ER(C_j)) \div ER(C_j)) \quad (5)$$

$$\sum_{i=1}^n mW(i) \quad (6)$$

- Once the first stump's performance is calculated, all the data points are reassigned with new weights if the total error $ER(C_j)$ is higher. The new weights represented by $nw(i)$ are calculated using the equation 7. The value of α in this equation is considered to be negative if the corresponding data point is classified correctly and positive if misclassified.

$$nw(i) = w(i) * e^{\pm\alpha} \quad (7)$$

- All the new weights are then normalized so that the total is equivalent to 1 using the equation 8.

$$nw(i) = nw(i) / \sum_{i=1}^n nw(i) \quad (8)$$

- Then all the above steps starting from step 1 to the previous step are repeated till the total error is reduced.

2) RANDOM UNDERSAMPLING AND BOOSTING ALGORITHM

Random UnderSampling and Boosting (RUSBoost) [33] is an algorithm that is an integration of both RUS and AdaBoost. RUS is used for randomly removing the data points from the majority classes and AdaBoost is used as an Ensemble technique. The steps of the algorithm [34] are as follows.

- Let us consider that 'M' is the data set and RUS will extract a random subset 'm(l)' from 'M' such that the size of m(l) is based on equation 9 where 'i' denotes the sampling size of the data set M.

$$2 * i < |M| \quad (9)$$

- The number of classes in 'M' is divided into majority class $I(m)$ and minority class $i(m)$.
- The AdaBoost methodology is used for building the ensemble model from this step onwards. Each subset $m(l)$ is used as the data set for every individual classification model $C(l)$ and learning is carried out where $l=0,1,2,\dots,L$. Here L denotes the total number of classifiers chosen for creating the ensemble model.

3) RANDOM SUBSPACE METHOD

Random Subspace Method [35] is generally called as "attribute bagging" or "feature bagging" and focuses on

reducing the correlation among the various estimators used in the ensemble training. This is achieved by training the classification model chosen on a random subset of features instead of taking the complete feature space. The initial subset chosen is considered to be a basis and training is performed. Then the different subset is taken for another model and training is performed. The following are the steps involved in the algorithm.

- Let us consider that the total number of training points be 'T' and the total number of features or attributes in the data set 'D' be 'F'.
- Let 'C' be the total number of classification models to be considered and 'i' be the individual models.
- Choose a subset 't(i)' of the 'T', so that $t(i) < T$. The $t(i)$ is maintained the same for all the individual 'i' models.
- A subset 'f(i)' of the 'F' is chosen for creating the training set for the classification model 'i'. The $f(i)$ set is chosen based on the replacement strategy for each model.

After the training is completed with the training set having $f(i)$ feature subset on the classification models 'C', the outputs of all the individual models 'i' are combined using either a voting scheme or any other basic ensemble technique to decide on the final result.

B. CLASSIFICATION LEARNING MODELS

This subsection discusses in detail the various classifiers as individual models for creating a strong ensemble learner. The classifiers discussed are Decision Trees (DTs) and k-Nearest Neighbour (KNN).

1) DECISION TREE

Decision Trees are a very popular ML algorithm that can be utilized for regression as well as classification problems [36]. This is a type of non-parametric supervised learning method for performing any kind of classification task. This follows a hierarchical tree structure. The tree includes a root node, branches and leaf nodes. The root node is the topmost node in the DT from where the dataset starts to divide into child nodes based on the various conditions. The child nodes that originate from the root node as a result of some decisions are called as decision nodes. If any specific node cannot be further split is called a leaf node. The process by which certain nodes in the DT are removed to prevent over-fitting is called as pruning. The algorithm of DT is as follows.

- The root node is the entire data set from where the algorithm begins.
- Having the root node, the best questions are asked based on the features available in the data set.
- Based on the answer obtained from the question, the root is further divided into two or more distinct groups.
- The above two steps are repeated until the tree lands with a leaf node where further split is not possible. These leaf nodes will specify the outcomes.

The DT classifier starts the process by making several assumptions so that it can land with effective models. A few assumptions made are as follows.

- 1) A recursive method of partitioning is followed. This process of splitting is continued until a stopping criteria is met.
- 2) The features using which the nodes are split are considered to be independent of each other.
- 3) A top-down greedy approach is utilized for constructing the tree. At each node, the split is done based on either the maximum information gain value or the minimum impurity measure.
- 4) The impurity measures can be either entropy or the Gini index.
- 5) Equal importance is given to all the features in the data set. Also, the data set do not have missing values and outliers.

The DT classifier uses the following measures for deciding the split at each node. The impurities at each node are usually calculated using entropy or Gini index. The entropy is used for calculating the uncertainty in the data set and helps in measuring the rate of disorder. Let D be the data set and S be the subset of D . Let $F(i)$ be the feature at a node where the impurity needs to be tested. Let 'P' be the probability of a positive class and 'p' be the probability of a negative class. Then the entropy for the feature $F(i)$ can be calculated using the equation 17.

$$E(F(i)) = [P(F(i)) * \log(P(F(i)))] - [p(F(i)) * \log(p(F(i)))] \quad (10)$$

After measuring the entropy at each node for a feature, when the entropy value is high then it represents that the impurity is high. This clearly states that the entropy is inversely proportional to purity. Once the entropy is used as a measure, then followed by this the information gain is also calculated for deciding on the split. Information gain specifies the level of reduction in uncertainty when a specific feature $F(i)$ is selected as the feature based on which the split is done. The information gain 'I' is measured using the equation 11.

$$I(S) = E(S) - E(S|F(i)) \quad (11)$$

The feature $F(i)$ which has the highest information gain value is selected for splitting the node further. These steps are continued until the stopping criteria is reached. The Gini Index is a measure that helps in identifying the possibility for mis-classification of an instance. The Gini Index GI can be calculated using the equation 12.

$$GI = 1 - \sum_{i=1}^n R(i)^2 \quad (12)$$

In the equation 12, $R(i)$ represents the ratio of the target class to the total number of observations for a particular node, 'n' specifies the target classes in the data set D . The Gini index GI is calculated for all the features at every node and the feature with lowest Gini index is taken as the criteria for split

of the root node. This process is continued at every level and proceeds until the tree lands with leaf nodes.

2) NEAREST NEIGHBOUR

The k-Nearest Neighbour algorithm is the simplest classifier [37]. This algorithm can be used for both numerical as well as categorical data. The algorithm does not make any initial assumptions. Also, kNN is not prone to be sensitive towards the outliers in the data set. The algorithm focuses on finding a data point that is nearer to the target data point selected using various kinds of distance measures. The distance metrics can be any one out of the following.

- 1) **Euclidean Distance:** This measures the straight line distance between two points. If (p_1, q_1) and (p_2, q_2) are the two points, then the Euclidean distance can be calculated using the equation 13 below.

$$ed(p_i, q_i) = \sqrt{(p_2 - p_1)^2 + (q_2 - q_1)^2 + \dots i} \quad (13)$$

- 2) **Manhattan Distance:** This measures the total distance that is travelled instead of the displacement between the data points. This distance can be calculated using the formula in the equation 14.

$$md(p, q) = \sum_{i=1}^n |p_i - q_i| \quad (14)$$

- 3) **Minkowski Distance** This distance metric focuses on the normed vector space. This seems to be a generalized form of the other two distance metrics discussed above. The measure is evaluated using the equation 15 where 'z' represents any integer between the two points considered.

$$mimd(p, q) = \sqrt[z]{\sum_{i=1}^n (p_i - q_i)^z} \quad (15)$$

The algorithm starts by fixing a k-value which usually represents the total number of nearest neighbours. Then the distance between the data points and the target point is calculated using any one of the distance metrics discussed above. Those k data points that have the lowest distance from the target point are grouped to be the nearest neighbours. Then the majority voting scheme is used to decide on the class label for the classification problem.

C. FEATURE SELECTION ALGORITHM

The feature selection algorithm used in the proposed methodology is the Maximum Relevance - Minimum Redundancy (MRMR) [38]. The features available in the data set are ranked based on the importance they have in predicting the final class. The importance is calculated based on its relevance and redundancy component. The relevance component is defined based on the correlation of the feature with its target class. If the target variable is categorical, then the relevance is based on the mutual information. If the target is continuous, then the relevance is the F-statistic score of the

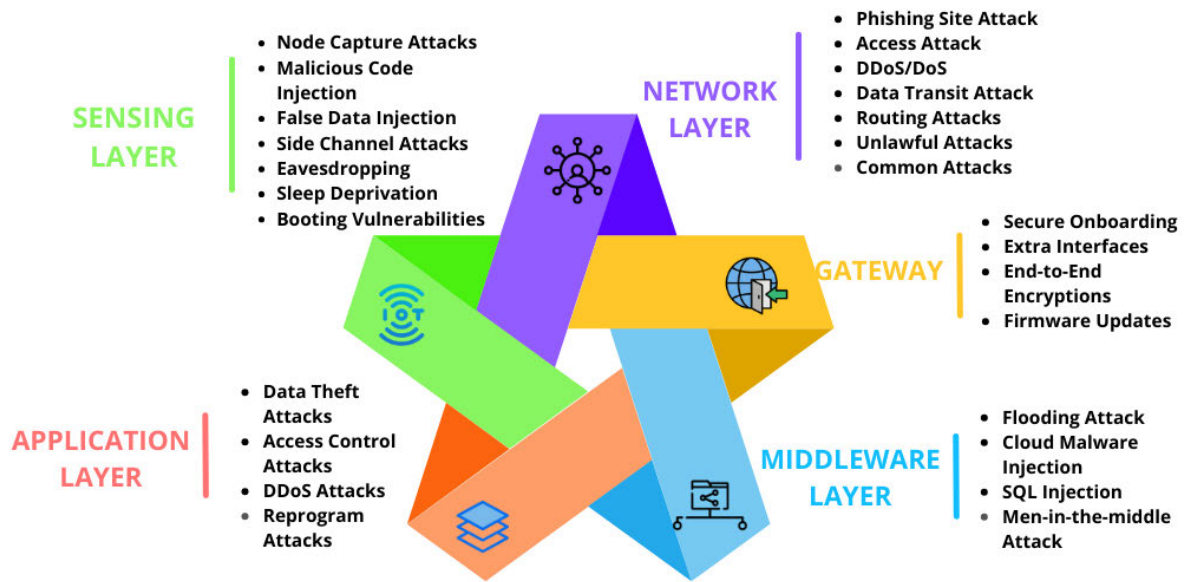


FIGURE 2. Types of IoT attacks - layer wise.

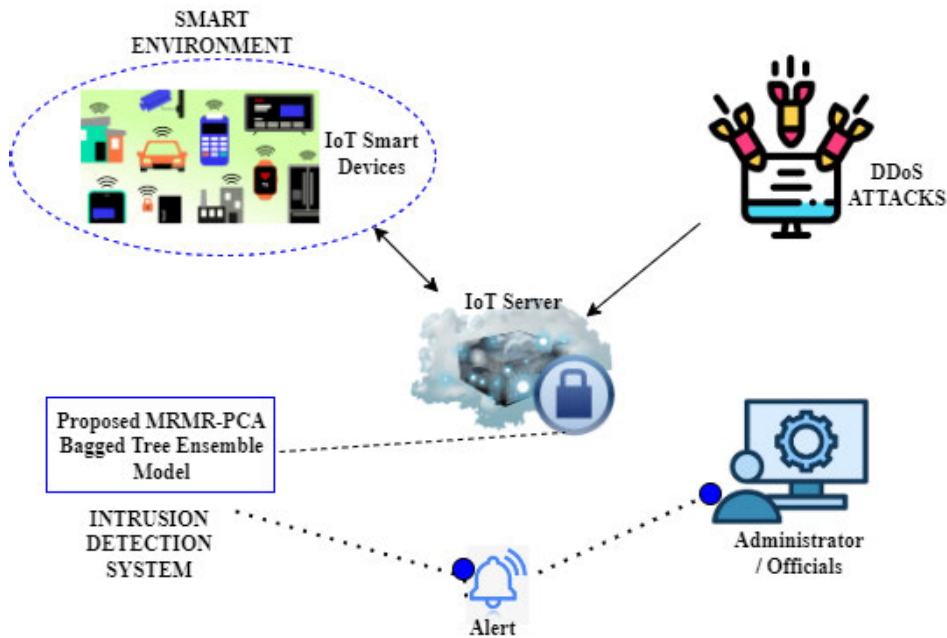


FIGURE 3. Real-time illustration of proposed framework.

feature with the target. The redundancy component specifies the correlation of the current feature with the already selected feature in the previous iteration. If the variable is a categorical one, then redundancy is based on its mutual information. If the variable is a continuous one, then the redundancy can be either its Pearson correlation coefficient or the distance between their vector space. This algorithm is best suited for data sets having features that are either fully categorical or numerical. The mutual information MI can be calculated using the equation 16 as shown below. Let us consider that the data set is represented as D , the feature is represented as

F , the Target is represented as T and entropy is represented as H .

$$MI(F) = H(F) - H(F|T) \quad (16)$$

The entropy H can be calculated using the equation 17 as shown below. Here, $p(T(i))$ represents the probability of the categorical target variable and ' i ' represents the number of possible categories in the target.

$$H(F) = - \sum_{i=1}^n p(T(i)) \log p(T(i)) \quad (17)$$

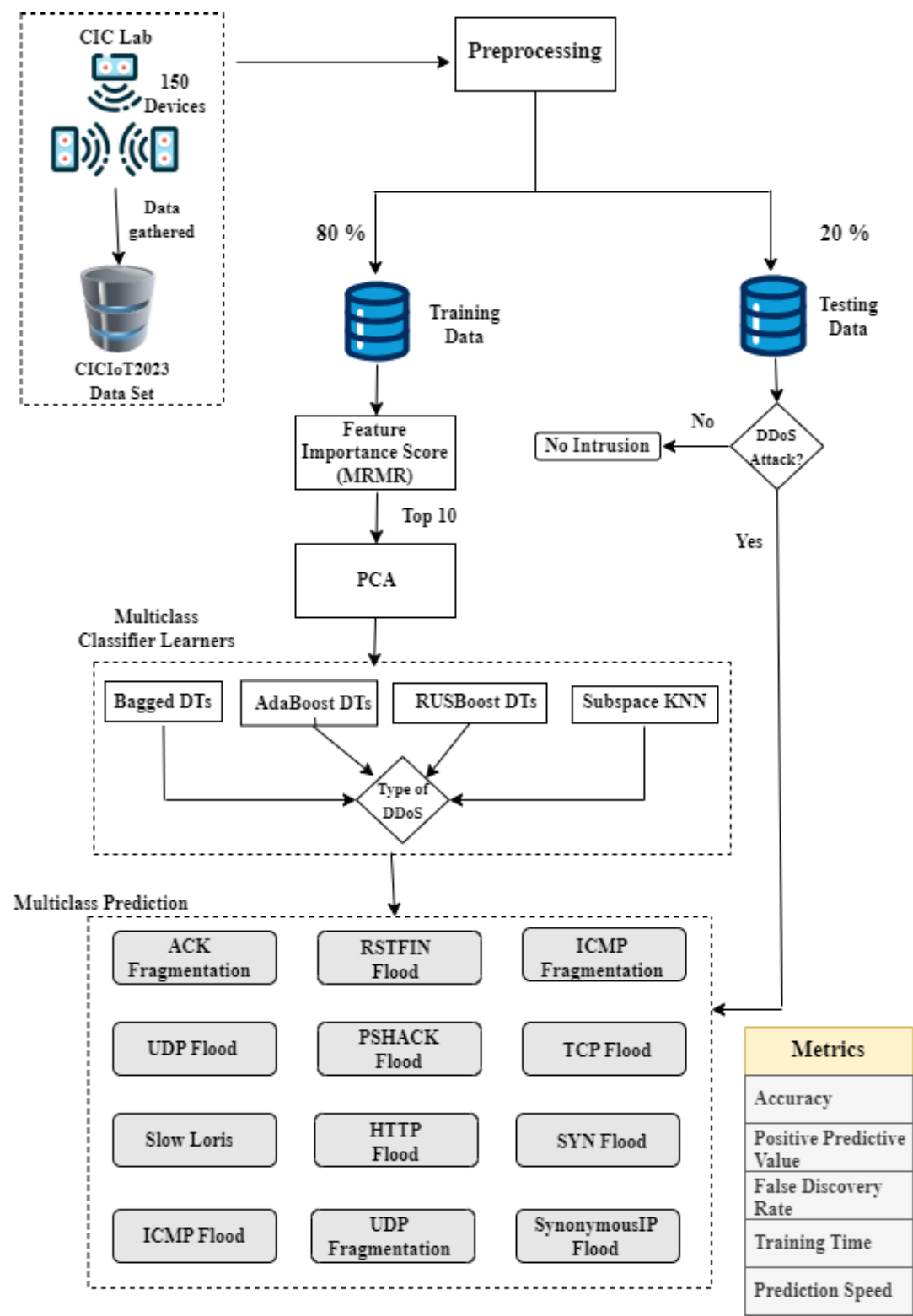


FIGURE 4. Proposed hybrid MRMR-PCA BagDT ensemble model.

The MI score is usually between 0 and infinity. If the score is high, then the correlation between the feature and target class variable is high and a 0 indicates a weak correlation between them.

V. PROPOSED METHODOLOGY

The IoT environment is usually made of different kinds of devices rapidly communicating among themselves. These devices are used for the exchange of valuable information

across boundaries to create value for human beings. IoT includes millions of devices interconnected via the Internet and they generate enormous amounts of valuable as well as sensitive data. The devices can be wearables, sensors, smart gadgets, computer terminals, home appliances or anything that can communicate. There are various sectors in which IoT devices are used like healthcare, agriculture, finance, retail, autonomous vehicles and industrial manufacturing. The IoT devices in all these sectors are really smart enough to send and receive data. The valuable data and information gathered are transmitted via the Internet and stored in an archive which can be a cloud storage. The devices do not have the smartness to detect if any attackers have intruded and whether the sensitive information is securely transmitted via the network. Various kinds of attacks are possible in an IoT environment and can occur by targeting different layers as illustrated in Figure 2. But the proposed model focuses on detecting the various attacks under the category DDoS attack namely ACKFragmentation, UDP Flood, SlowLoris, ICMP Flood, RSTFIN Flood, PSHACK Flood, HTTP Flood, UDP Fragmentation, ICMP Fragmentation, TCP Flood, SYN Flood and SynonymousIP Flood.

The proposed framework model is based on the Ensemble approach. An Ensemble-based classification learner model is proposed which detects the attack type of DDoS attack category. The framework for the IoT environment consists of three components as illustrated in Figure 3. The first component is the set of IoT devices in various sectors, the second component is the intruders who try to steal or misuse the sensitive data gathered by the devices and the third component is the Intrusion Detection System which detects the attacks. Let us consider that there are various IoT devices like smart gloves, smart pacemaker, smart wristbands, and smart jackets in the healthcare sector, smart drones, smart water dispensers, smart pest detectors concerning the agriculture sector and smart Sensex rater to finance sector. These smart devices are interconnected via the internet and the data is being gathered and transferred via the communication channel to the storage area like a cloud server. In the cloud, based on the archived data, sensitive decisions might be taken. If there are attackers and intruders in between the communication channels, the sensitive data might be misused or modified before being archived in the cloud server. Due to this, there is a chance for false alarms or false decisions which might even lead to serious health issues or marketing strategies or wrong decisions in agriculture. To overcome these issues, the proposed framework has an Intrusion detection system which is discussed later in this section. The proposed model focuses on the design of this intrusion detection system. The proposed model is validated using the universal benchmark CIC IoT 2023 (CIC23) data set. The proposed intrusion detection system consists of various stages as shown in Figure 4. The preprocessed intrusion detection data set is first fed into the system. The features are ranked using the MRMR feature selection algorithm. The top ten features are selected based on the

importance score. Then the traditional PCA method is applied to the selected features to verify if there is any further possibility in reduction of dimension. Bagged Decision Trees are used as the learners for model creation. The resultant Hybrid MRMR-PCA BagDT model is the strong ensemble learner that can handle huge data in a short period and the algorithm is shown in the Algorithm 1. The most robust model that can be adapted in real-time is the Bagging-based Decision Tree Ensemble model as it is a lightweight model.

Algorithm 1 Proposed Hybrid MRMR-PCA BagDT Ensemble Model

Input: Labelled Dataset D from CICIOT2023

Output: Labelled Class $L_i(A)$, $i \leftarrow 1, 2, \dots, l$, $l \leftarrow 34$,
 $A \leftarrow attacks$

begin

 Create $R \subset D$, such that $\forall (L_i(A)) \in A \leftarrow DDoS$

 Create $T_r \subset R$ and $T_l \subset R$ such that $T_r \leftarrow 0.8 * R$
 and $T_l \leftarrow 0.2 * R$

while T_r **do**

 Calculate MI, $\forall$ features F_i , $i \leftarrow 46$

$F_f(j) \leftarrow Rank\ F\ using\ MI(F_i)$, $j \leftarrow 1, 2, \dots, 34$

$F_f(k) \leftarrow k = 1, 2, \dots, 10$

 Apply PCA on $F_f(k)$ to explain variance of 95%

 Create new training datasets $T_{r1}, T_{r2}, \dots, T_m$
 such that $n \leftarrow 30$

for $n \leftarrow 1$ to 30 **do**

 Train each dataset T_m

 Build classifier model C_n

while $T_l(m)$ **do**

for $i \leftarrow 1, 2, \dots, m$ **do**

$L_i(T_l(m_n)) \leftarrow Classify\ each\ instance\ m\ of\ T_l$
 using classifiers C_n

$L_i(T_l(m)) \leftarrow averaging\ scheme$

VI. RESULTS AND DISCUSSION

This section discusses in detail the description of the data set CICIOT2023 (CIC23) used in the experimental evaluation, the results of the feature selection technique applied for extracting the top features, further reduction in feature space using PCA, the classification models used and their results.

A. DATASET DESCRIPTION

The dataset which is used in this research is CICIOT2023 [25]. This dataset is a real-time data set that is produced in the Canadian Institute of Cybersecurity (CIC) Lab using 105 IoT devices. Out of these 105 devices, 67 devices are directly involved which mimics a smart home environment and the remaining 38 are interconnected to this environment via hubs. The data set is created by generating different types of attacks namely DDoS, DoS, Recon, Web-Based, Brute Force, Spoofing and Mirai using various tools and devices in the IoT environment. Our research focuses on developing a robust ensemble model for detecting the types of DDoS attacks. Hence the data set is made ready for our requirement

TABLE 5. DDoS - attack types and instances.

Attack Type	Observations /Instances	Framework for Attack Execution	Description
ACK Fragmentation	2,85,104	hping3 [39]	Maximum number of small sized packets are used to slow down the network
UDP Flood	54,12,287	udp-flood [25]	The target device is attacked by flooding numerous number of packets which are of different types.
ICMP Flood	72,00,504	hping3 [39]	
HTTP Flood	28,790	golang-httpflood [25]	
TCP Flood	44,97,667	hping3 [39]	
SlowLoris	23,426	slowloris [25]	A web server is targeted using HTTP requests
RSTFIN Flood	40,45,285	hping3 [39]	Numerous RST-FIN packets are forwarded to degrade the network
PSHACK Flood	40,94,755	hping3 [39]	A server operation is degraded by forwarding ACK and PUSH packets
UDP Fragmentation	2,86,925	udp-flood [25]	This is a special type of UDP flood which consumes high bandwidth
ICMP Fragmentation	4,52,489	hping3 [39]	Numerous identical fragmented ICMP messages are used
SYN Flood	40,59,190	hping3 [39]	A large number of synchronize packets are sent to the server but it also takes care that the handshake is not complete
SynonymousIP Flood	35,98,138	hping3 [39]	Manipulated TCP and SYN packets are forwarded to utilize the server's resources

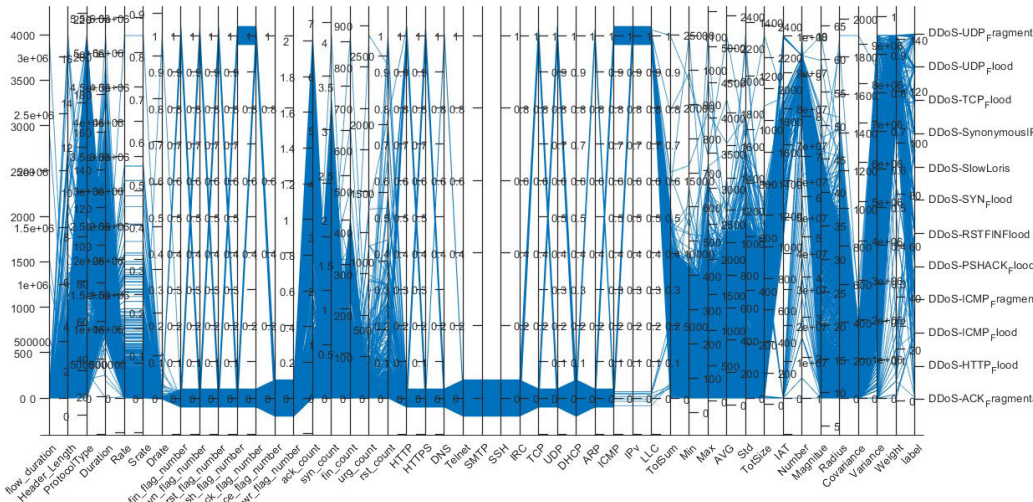


FIGURE 5. Dataset visualization.

as shown in the table 5 below. The DDoS attack can be of any type among the twelve types listed in the table. The data set consists of totally 46 features namely flow duration, Header Length, Protocol Type, Duration, Rate, Srate, Drate, fin flag number, syn flag number, rst flag number, psh flag number, ack flag number, ece flag number, cwr flag number, ack count, syn count, fin count, urg count, rst count, HTTP, DNS, Telnet, SMTP, SSH, IRC, TCP, UDP, DHCP, ARP, ICMP, IPv, LLC, Tot sum, Min, Max, AVG, Std, Tot Size, IAT, Number, Magnitude, Radius, Covariance, Variance and Weight. All these features have numerical values.

B. PREPROCESSING

The step preprocessing of the dataset includes steps like investigating the features by visualizing the features using

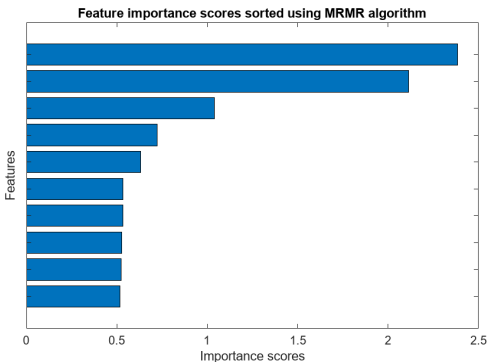
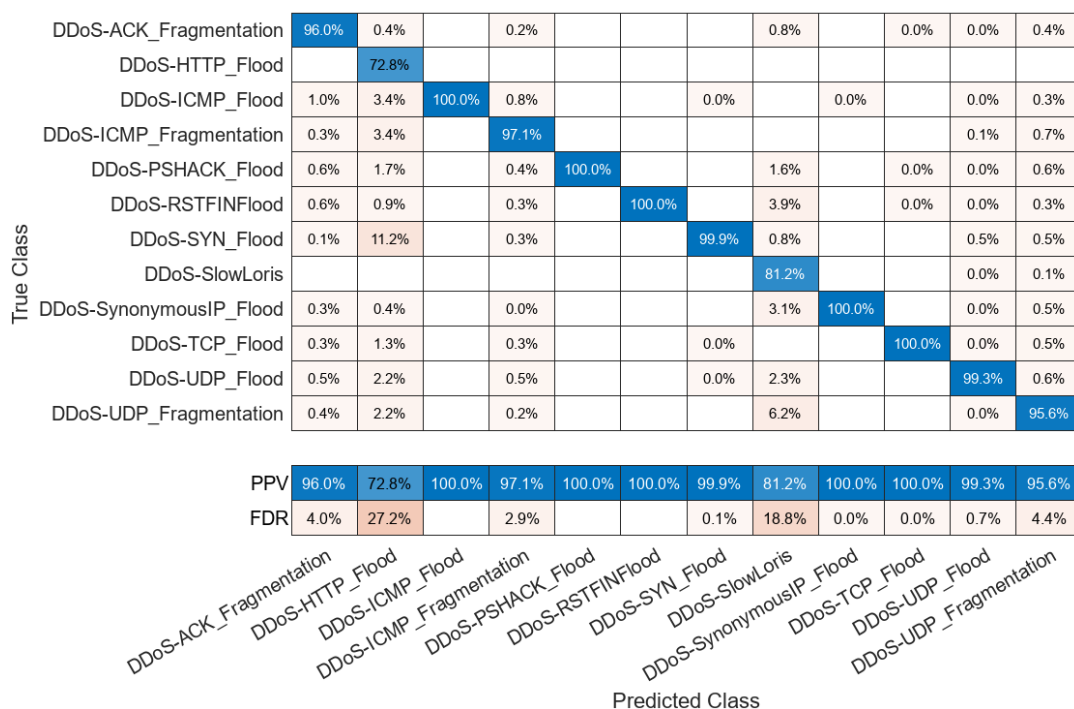
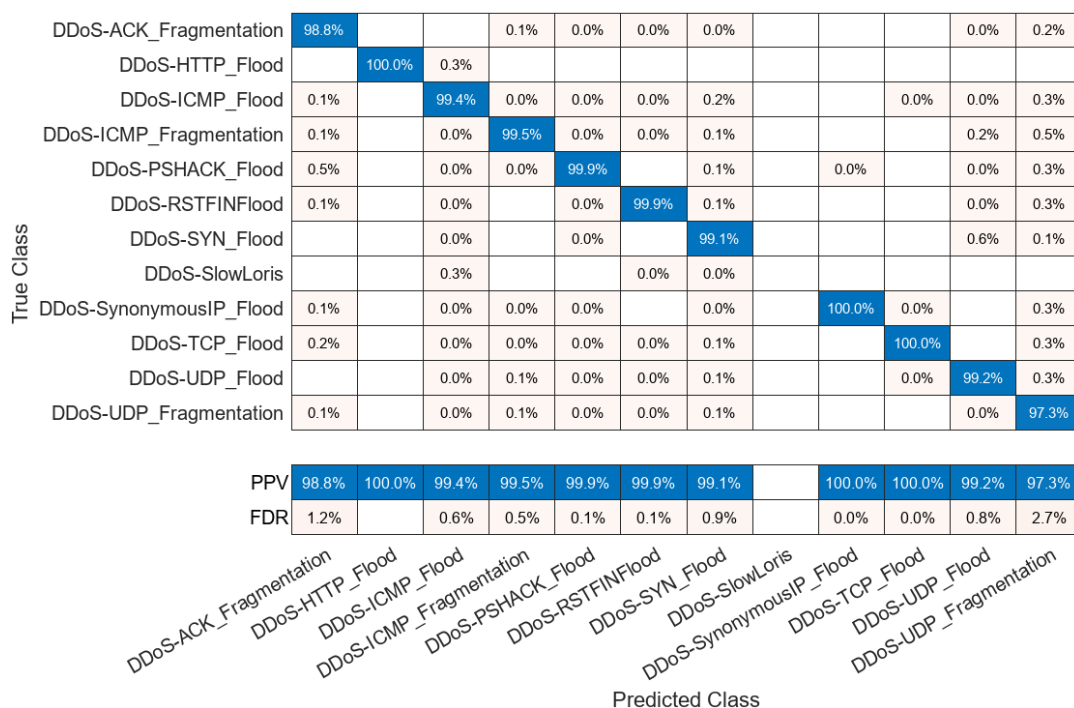


FIGURE 6. Features extracted using MRMR algorithm.

scatter plots, parallel plots and bar plots. From the visualization, it was noticed that the dataset consisted of 47 features out



type float. Then the dataset was investigated to check if any missing values are available. There were no missing values in the dataset. Then, the dataset was investigated to check for availability of outliers and no outliers were present. The

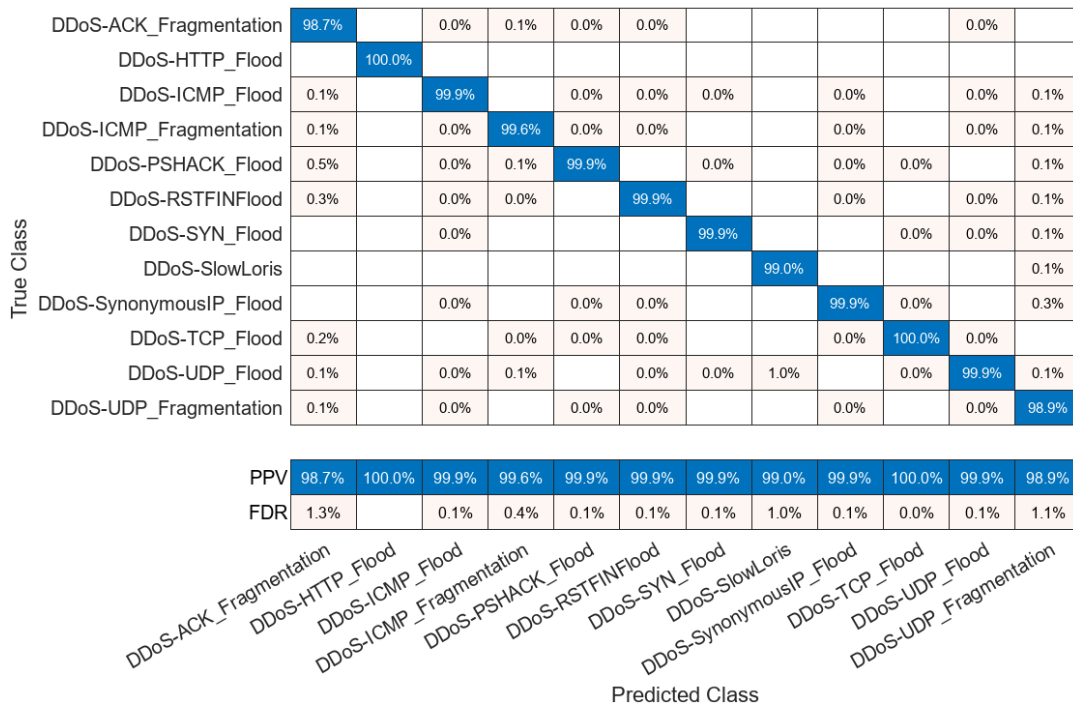


FIGURE 9. Confusion matrix-hybrid MRMR-PCA BagDT ensemble model.

TABLE 6. Importance score for features using MRMR.

Number	Feature	Importance Score
1	IAT	2.3829
2	ARP	2.1133
3	SSH	1.0390
4	DNS	0.7218
5	Drate	0.6302
6	syn_flag_number	0.5337
7	ack_count	0.5333
8	psh_flag_number	0.5264
9	ICMP	0.5241
10	UDP	0.5161
11	Weight	0.4938
12	Std	0.4469
13	TCP	0.4408
14	flow_duration	0.3875
15	rst_flag_number	0.3798
16	ack_flag_number	0.3765
17	Min	0.3055
18	fin_flag_number	0.2861
19	syn_count	0.2605
20	HTTP	0.2413
21	urg_count	0.2197
22	Rate	0.2189
23	fin_count	0.2189
24	ProtocolType	0.2092
25	Header_Length	0.2001

data in each of the 46 features are then normalized using standard scaler Z-score since all the features were of floating point type. The Z-score type used is Standard Deviation. The preprocessed data is then subdivided into training dataset and testing dataset based on the 80-20 rule. 80% of the data is

considered for training and 20% for testing and validation purpose.

C. FEATURE SELECTION

The original data set CICIoT2023(CIC23) as discussed in the previous section has totally 46 features. When all these features are used for learning, the total cost would be higher and the prediction speed would also be affected. Hence MRMR algorithm discussed in previous section is used for ranking the features based on their importance score. The importance score for 25 features out of total 46 are tabulated in the table 6. Based on the importance score, the features are ranked and the top 10 features are alone extracted as shown in the figure 6 below.

D. PRINCIPAL COMPONENT ANALYSIS

Though top 10 features are extracted using the previous step, in order to increase the prediction speed and reduce the total cost as well as the training time required, the traditional PCA method is used for still reducing the dimension of the feature space. The PCA is configured here to maintain the features so that at-least 95% variance is explained. When this is applied to the extracted feature space, PCA maintained only one component out of the 10 features extracted. The explained variance per component is tabulated in the table 7 below.

E. CLASSIFICATION MODELS

The Ensemble Learning methods namely AdaBoost, RUS-Boost, Random Subspace and Bagging are applied on

TABLE 7. Explained variance - PCA.

Component	IAT	ARP	SSH	DNS	Drate	syn_flag_number	ack_count	psh_flag_number	ICMP	UDP
Explained Variance (%)	100	0	0	0	0	0	0	0	0	0

TABLE 8. Performance of proposed MRMR-PCA hybrid ensemble models.

Model	Training Time (secs)			Prediction Speed (observations per sec)			Accuracy (%)
	All Features	MRMR	MRMR-PCA hybrid	All Features	MRMR	MRMR-PCA hybrid	
AdaBoost DT	399.17	86.661	48.273	29,000	40,000	42,000	99.6
RUSBoost DT	205.33	105.49	56.368	22,000	28,000	41,000	99.7
Bagged DT	436.47	93.594	60.817	29,000	40,000	43,000	99.9
Random Subspace KNN	4843.1	591.44	78.93	180	1200	23,000	99.9

individual traditional classification models like Decision Trees and KNN. A 5-fold cross validation is performed for validating the final robust ensemble model. The performance of the models are evaluated without feature selection, with feature selection and with feature selection followed by PCA as tabulated in table 8.

1) ADABOOST BASED DECISION TREE MODEL

The ensemble algorithm AdaBoost is utilized for building a robust classification model using the traditional Decision Tree method. The maximum number of splits allowed is fixed as 20, total number of learners is initialized as 30 and the Learning rate is 0.1. When the training is performed and validated using the 5-fold cross validation, the accuracy obtained is 99.6%, prediction speed is approximately equal to 42,000 observations per second and the time required for training is 48.273 seconds. Once the model is deployed, the size of the model would be approximately equivalent to 1 MB. The confusion matrix shown in the Figure 7 specifies that when the AdaBoost DT Model is used, though the overall accuracy obtained is 99.6%, the individual class “SlowLoris” is completely misclassified. Hence this model cannot be used in real-time for IDS.

2) RUSBOOST BASED DECISION TREE MODEL

The traditional Decision Tree learner is now ensembled using the RUSBoost algorithm with hyper-parameters configured as number of splits equivalent to 20, number of learners equivalent to 30 and learning rate as 0.1. As a result of the training and validation the accuracy is 99.7%, speed of prediction is 41,000 observations per second and the time required for training is 56.368 seconds. The confusion matrix in the Figure 8 clearly specifies that the False Discovery Rate (FDR) for two classes namely DDoS-HTTP-Flood and DDoS-SlowLoris are 27.2% and 18.8% respectively. Hence, this model can also increase the mis-classification rate when used in real-time.

3) BAGGING BASED DECISION TREE MODEL

The traditional classifier Decision Tree methodology is used to create a robust ensemble model based on the bagging technique. The model hyper-parameters are as follows. The maximum number of splits allowed are fixed as 238686. The total number of learners is fixed to 30. The total number of predictors allowed for sampling is 46. The experimental results show an accuracy of 99.9%, prediction speed of 43,000 observations per second and training time of 60.817 seconds for a total of 2,38,687 observations. The confusion matrix based on 5-fold cross validation shown in the Figure 9 clearly depicts that the positive predicted value rate is high for all the twelve classes of DDoS attack category and false discovery rates are very low.

4) RANDOM SUBSPACE BASED KNN MODEL

The Random Subspace Approach of ensemble is utilized for performing learning using the classification learner K-Nearest Neighbour. The subspace dimension is configured to be 5 and the total number of learners is 30. The accuracy as a result of learning is 99.9% at a prediction speed of 23,000 observations per second. The total time required for training the observations of 2,38,687 is 78.93 seconds. During deployment, the model would require a size of 200 MB. The confusion matrix after 5 fold cross validation is shown in the Figure 10. When this ensemble model is used, the Positive Predictive Value (PPV) for all classes are high but the model size is around 200 MB. Due to this huge size, this model cannot be deployed in real-time IoT environment as it is made of light weight devices. Also the prediction speed is comparatively low.

F. DISCUSSION ON RESULTS

The experimental analysis clearly specifies that our proposed Ensemble based MRMR-PCA framework for classifying the DDoS attacks in real-time IoT environment outperforms the state of the art ML classifiers like Decision Tree and KNN. The proposed framework also shows better performance

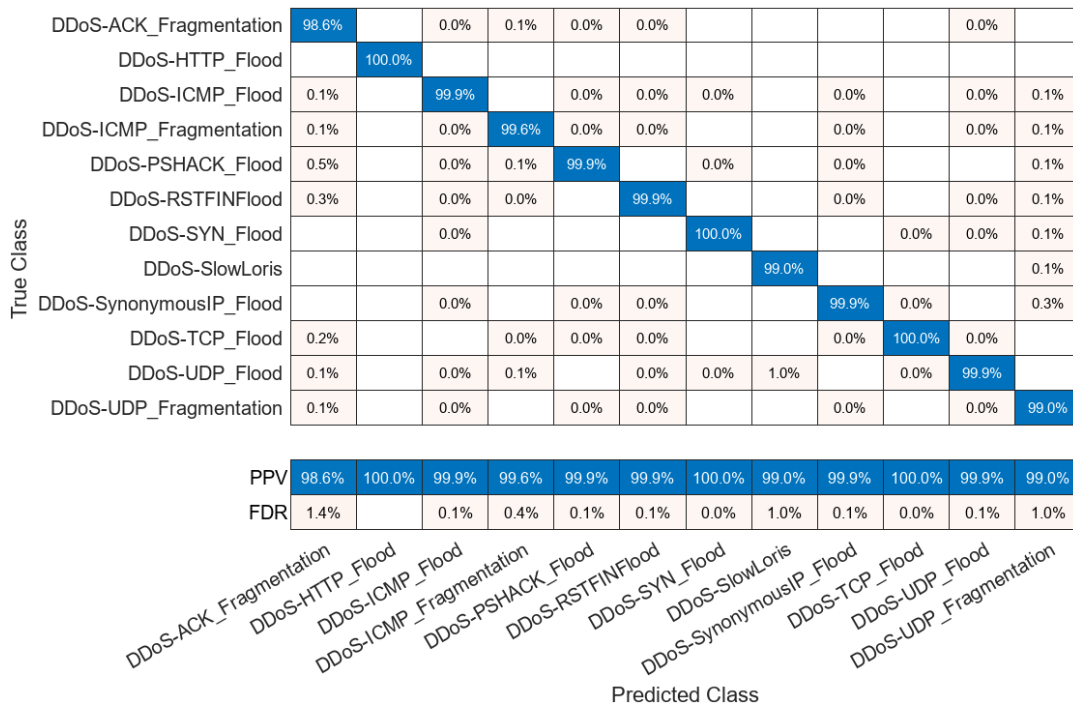


FIGURE 10. Confusion Matrix-Hybrid MRMR-PCA SubspaceKNN Ensemble Model.

TABLE 9. Performance analysis-proposed model vs state-of-the-art techniques.

Ref.	Feature Selection Included	Dimension Reduction Included	Class Imbalance Issues Solved	Lightweight Model	Overfitting Issues Handled	Accuracy %
[1]	×	×	×	×	×	98.75
[2]	×	✓	✓	✓	×	93.13
[3]	✓	×	×	×	×	99.51
[4]	×	×	×	✓	×	97.24
[5]	×	✓	×	×	✓	99.2
Proposed	✓	✓	✓	✓	✓	99.9

than the traditional Ensemble based classification model of Decision Tree and KNN. From the table 9 it is evident that our proposed model outperforms the other state-of-the-art techniques that are implemented and tested on the CICIoT2023 dataset by other researchers. The reason behind the improved performance is as follows.

- The MRMR acts as a feature measurement filter based on mutual information among the features and identifies the most relevant as well as non-redundant subset of features which improves the model's learning accuracy.
- PCA prevents over-fitting when a huge amount of data is being used for learning. Also when the feature space is still reduced, the training time is reduced drastically.
- Ensemble learning approach reduces the variance and makes the final resultant model more balanced with less bias as well as variance. The resulting ensemble model is always considered to be a very strong learner.
- The proposed hybrid framework extracts the highly important feature and excludes the others in the features space hence increasing the accuracy of the resultant strong learner.

The contributions of this proposed framework in IoT environment for Intrusion Detection are as follows.

- The proposed methodology which combines the MRMR feature ranking algorithm and traditional PCA extracts the predominant features from the entire feature space of the data set. The reason behind the improved performance is that two levels of extraction and filtering are performed. Initial filtering is done using MRMR and then PCA is used to calculate the variance among the filtered features. This still reduces the feature vector space hence increasing the prediction speed and accuracy. Also due to the reduction in size of feature space, training time is reduced.
- The percentage of reduction in the dimensionality of the feature space is higher when the proposed hybrid MRMR-PCA methodology is used.
- The training time required for the learner is reduced drastically.
- The classification accuracy is improved when the proposed framework is used compared to traditional models.

- The prediction speed is increased when the proposed hybrid approach is used.
- The model size is compact which paves way for easy real-time deployment.
- The proposed MRMR-PCA based Bagged Decision Tree Ensemble Model and MRMR-PCA based Random Subspace KNN Ensemble Model can be effectively used in real-time IoT environments in all sectors as discussed in the previous sections. The major reason is that these models can efficiently handle huge data from IoT devices and can alert officials faster when any intrusion happens.

VII. CONCLUSION

This paper proposes a hybrid MRMR-PCA based Bagged Decision Tree Ensemble Model for the detection of the DDoS attack types in real-time IoT environment. The hybrid model is validated using a benchmark data set CICIoT2023 which has twelve attack types for the DDoS category. The proposed framework classifies all the twelve attack types with an accuracy of 99.9% at a prediction speed of 43,000 observations per second. The training time required is 60.817 seconds. The model size is also compact enough approximately equal to 2 MB, so that it can be deployed easily in real-time IoT environment. Since the accuracy and prediction speed is high, when any intrusion occurs in the environment, it can be detected faster and alerts can be passed over to the concerned officials. Since the EL approach is used, the model developed is a very strong learner and can adapt itself dynamically to the environment. It also supports self learning in due course of time. The same hybrid model is not yet tested for detecting other types of attacks in IoT environment which is the path for future researchers.

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